

Report for Image Quality vs Computer Vision

CE6461 – Computer Vision Systems

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Contents

Introduction	2
Results Table.....	2
Visual Evidence	3
Correlation SFR50 with confidence:.....	5
Strength of ssdmobilenetv2; defocusing and sharpening tolerance	5
SFR50 minimum to be recognized	5
Vital reason to control the focus and sharpness.....	6
Conclution	6
References	6
 Figure 1: MTF Plot (In-focus).....	 3
Figure 2: MTF Plot (Strong-defocus).....	4
Figure 3:Object Detection.....	4

Introduction:

The study questions of this lab are the following: How does the quality of camera images affect downstream object detection? The study combines a quantitative sharpness signal (SFR50 of the slanted-edge method) with ssdmobilenetv2 confidence scores on images taken under controlled focus and sharpening conditions with a Raspberry Pi camera. The experiment introduces a direct, empirical relationship between optical / INSP and model behavior and three sharpening conditions at in-focus, mild-defocus and strong-defocus for chart-image-based decisions and confidence-based object-image-based decisions through the measurement of SFR50, and a confidence of model behavior at three sharpening conditions at in-focus, mild-defocus, strong-defocus.

Results Table:

Focus Setting:	Sharpness:	SFR50 (cy/px):	Confidence Score:
	1-Default	0.177	Mouse (59%)
In-Focus	0	0.122	Mouse (64%)

Image Quality vs Computer Vision

	8	0.374	Mouse (0%)
	1-Default	0.0384	Mouse (73%)
Mild Defocus	0	0.0371	Mouse (69%)
	8	0.0508	Mouse (65%)
	1-Default	0.0243	Mouse (66%)
Strong-Defocus	0	0.025	Mouse (68%)
	8	0.026	Mouse (53%)

Visual Evidence:

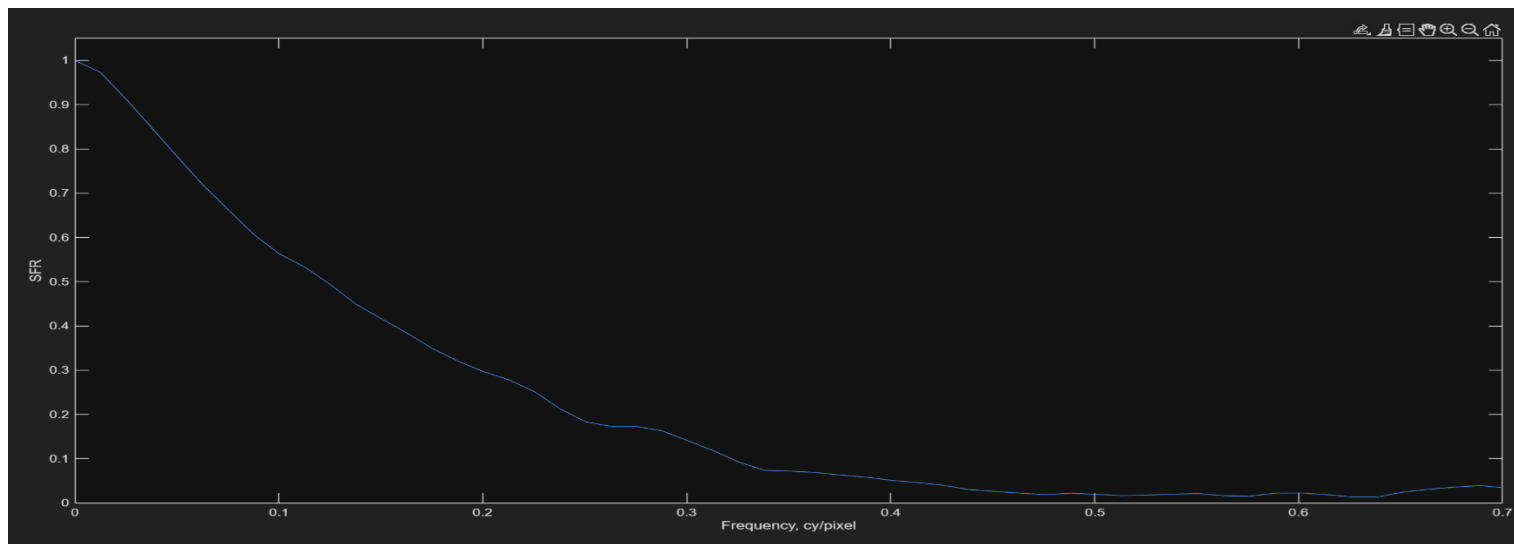


Figure 1: MTF Plot (In-focus)

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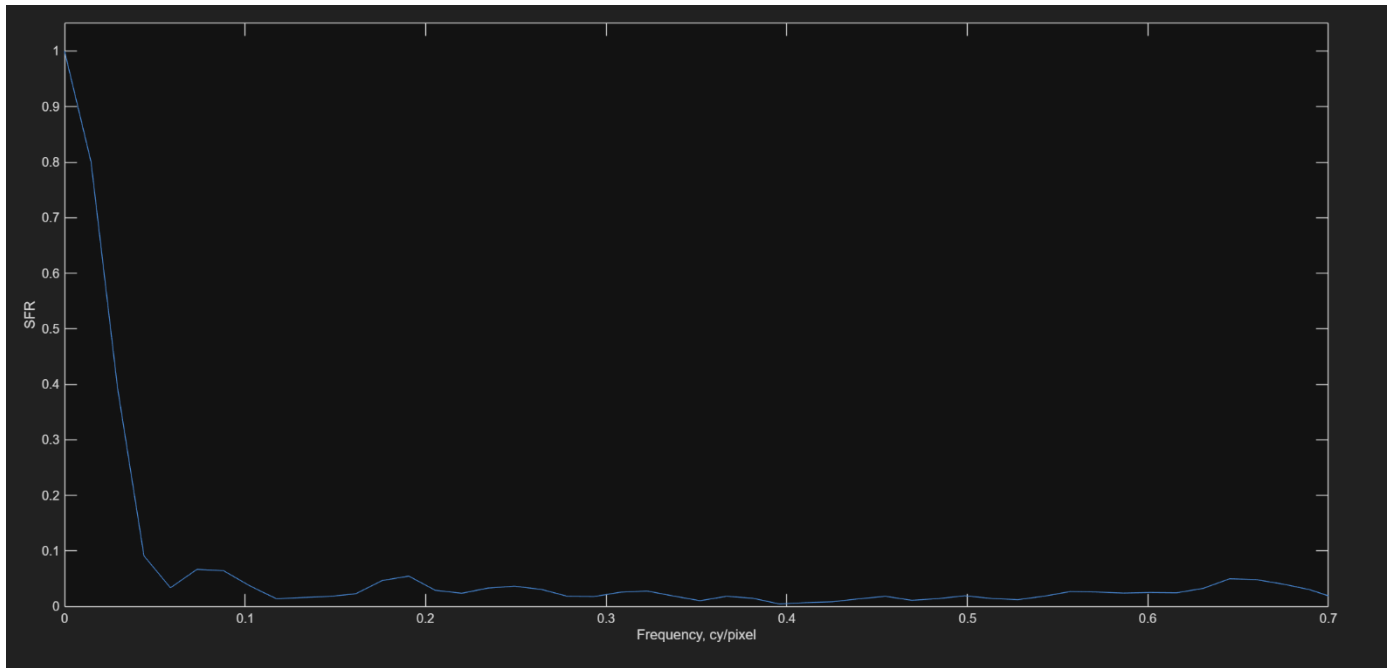


Figure 2: MTF Plot (Strong-defocus)



Figure 3: Object Detection

Correlation SFR50 with confidence:

In the nine conditions, confidence remains relatively steady between 53-73% with SFR50 range being extensive and varies between 0.024 to 0.177 and hence the total correlation is weak and even negative.

The most apparent outlier is the "In-Focus, sharpness 8" because SFR50 is most intricate at 0.374 and the detector gives 0% confidence, indicating that a high SFR50 that is forced by sharpness aggressive sharpening may be accompanied by failure.

With the decrease of focus to in-focus to mild and strong defocus, the performance does not change substantially (i.e. the default in the mild-defocus is 73, in the strong-defocus is 66) and the performance is relatively unchanged by blur in this configuration.

The probably reason is because moderate blur causes the network upon which the model is based to be kept, and strong sharpening causes halos/ringing that distorts edge statistics that the model predicts, which explains the in-focus sharpness-8 failure despite an exceedingly high SFR50.

Strength of ssdmobilenetv2; defocusing and sharpening tolerance:

The model can still operate with SFR50 = 0.024-0.026 in the strong-defocus rows still giving 53-68% confidence, hence significant tolerance to defocus of this object and distance.

It solves the default and minimum sharpening during focus settings, yet in the in-focus example the maximum sharpening (SFR50 0.374) results in miss at 0% confidence showing sensitivity to over-sharpening artifacts on already sharp image.

Even sharpness 8 is still detected under defocus (65% mild, 53% strong) which means that the model can be more susceptible to over-sharpening the base image rather than blurring out the artifacts.

SFR50 minimum to be recognized

With $\geq 50\%$ confidence, the lowest SFR50 that passed this test in your data is approximately 0.025 cycles/pixel since all measurements that were strong-defocus and at 0.024-0.026 were above this threshold were at 50% confidence.

Given that no results less than $= 0.024$ were obtained, $= 0.025$ is a safe value at which SFR50 can be considered in this camera-object-model set-up, provided that sharpening is not drastic.

Vital reason to control the focus and sharpness:

What you have found is that SFR50 on its own is no predictor of the quality of the recognition because extreme sharpening can increase SFR50 but destroy detection, and means of ISP have to be tuned instead of being tuned to sharpness.

A consistent focus which keeps SFR50 above ≈ 0.025 and no vigorous sharpening eliminates distribution shift, edge statistics the model was trained on, and avoids artifact-driven failures such as the in-focus sharpness-8 miss.

Conclusion:

It is demonstrated that object-detection confidence does not vary greatly between broad changes in SFR50 caused by defocus, whereas aggressive sharpening capable of increasing SFR50 but with a negative or even negative impact on detection, thus indicating that increased sharpness is not always a desirable property on learned features. In this configuration the detector was stable at SFR50 values of around 0.025 cy/px, indicating a practical lower limit of focus quality and indicating deployment control to moderate and consistent sharpening and SFR50 at or above this threshold to avoid failure of the system in practice by artifact causes.

References:

- 1) CS6461 Lab-6: Image Quality vs Computer Vision